

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2011

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?

- A. Plasma
- B. White blood cells
- C. Red blood cells
- D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C]
[D] 24. [A] [B] [C] [D] 34. [A]
[B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2011.

1. Which of the following diseases is a child immunized against at birth, sixth week, tenth week, and fourteenth week after birth?
 - A. Measles.
 - B. Yellow fever.
 - C. Tuberculosis.
 - D. Poliomyelitis.
2. Which one of the following is advisable when spraying with chemicals?
 - A. Spraying when the temperature is very high.
 - B. Wearing gas mask when spraying.
 - C. Spraying against the direction of the wind.
 - D. Washing hands before spraying.
3. Which one of the following pairs consists only of omnivorous animals?
 - A. Chimpanzee and pig.
 - B. Chimpanzee and hyena.
 - C. Hippopotamus and crocodile.
 - D. Hippopotamus and pig.
4. The reasons why some plants have thick waxy cuticle is to
 - A. Store manufactured food.
 - B. Reduce water loss.
 - C. Allow breathing to take place.
 - D. Trap more sunlight.
5. Which one of the following consists of groups of food that mainly contains fats?
 - A. Avocado, groundnuts, coconut.
 - B. Maize, coconut, beans.
 - C. Irish potatoes, groundnuts, maize.
 - D. Irish potatoes, beans, avocado.
6. The first step in obtaining salt solution and pieces of copper wire is
 - A. Filtering.
 - B. Winnowing.
 - C. Using magnet.
 - D. Evaporation to dryness.
7. Substances that have definite mass and volume are
 - A. Solids only.
 - B. Solids and liquids.
 - C. Liquids only.
 - D. Liquids and gases.
8. A lactating mother should feed on a diet rich in calcium and phosphorus in order to
 - A. Enable her regain strength.
 - B. Replace blood lost during birth.
 - C. Improve the immunity of her baby.
 - D. Strengthen the bones of the baby.
9. Which one of the following is the most effective preventive measure of sexually transmitted infections among the youth in schools?
 - A. Seeking medical attention.
 - B. Use of condoms.
 - C. Circumcision.
 - D. Abstinence.
10. Which one of the following pairs consists only of inclined planes?
 - A. Wheel barrow and see-saw
 - B. Road winding up a hill and staircase
 - C. Ladder and see-saw
 - D. Wheel barrow and staircase
11. The percentage of the component of air in the atmosphere used by legumes to make proteins is
 - A. 78%
 - B. 21%
 - C. 0.97%
 - D. 0.03%
12. The following are signs and symptoms of anaemia EXCEPT
 - A. Breathlessness.
 - B. Pale white eyes.
 - C. Muscle wastage.
 - D. Tiredness.
13. Which one of the following consists only of a pair of animals that have constant body temperature?

- A. Shark and whale.
- B. Tortoise and ostrich.
- C. Shark and bat.
- D. Bat and ostrich.

14. The following are some activities carried out in an experiment to investigate drainage in soil

- i) Put equal amount of three type of soil in separate funnels.
- ii) Place the funnels on the mouth of each of the three plastic bottle.
- iii) Put cotton wool in the neck of each funnel.
- iv) Pour equal amount of water on each type of water on each type of soil at the same time.

Which one of the following is the correct order of the activities?

- A. (i) (ii) (iii) (iv)
- B. (iii) (ii) (i) (iv)
- C. (i) (iii) (ii) (iv)
- D. (ii) (i) (iii) (iv)

15. Which one of the following methods of soil conservation can be used to control splash erosion?

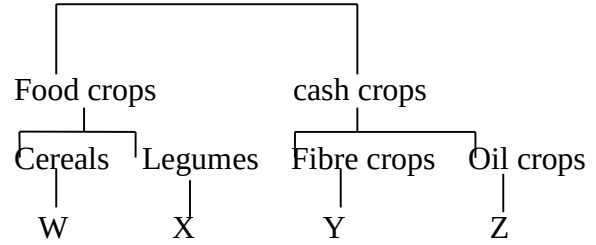
- A. Making terraces.
- B. Building gabions.
- C. Planting cover crop.
- D. Practicing contour farming.

16. The effects of heat on ice that make it change to gaseous form are

- A. Freezing and condensation.
- B. Melting and evaporation.
- C. Condensation and evaporation.
- D. Freezing and melting.

17. The chart below represents a simple classification of crop

Crops



Which one of the following crops are **CORRECTLY** represented by **W, X, Y** and **Z**?

W X Y Z

- A. Sorghum peas sisal sunflower
- B. Peas wheat sunflower cotton
- C. Green gram wheat sisal pawpaw
- D. Rice green gram kales cotton

18. The following is a feeding relationship that was observed in a certain feeding habitat

Maize grains → rats → K → hawks

Which one of the following is correctly represented by K?

- A. Frogs.
- B. Snakes.
- C. Weaverbirds.
- D. Lizards.

19. Which one of the following pests is **CORRECTLY** matched with the part of the crop attacked?

Pest	part of crop attacked
A. Aphids	grains.
B. Stalkborers	leaves.
C. Cutworm	stems.
D. Weevils	leaves.

20. Which one of the following diagrams represents what is observed when water is poured into a tin can with holes of the same size at different depth?

DIAGRAM

21. The diagram below represents a thermometer used to measure temperature of a certain liquid.

DIAGRAM

From the thermometer, the temperature of the liquid is

- A. 30°C
- B. 27°C
- C. 20°
- D. 10°

22. The diagram below represents a set up that was to demonstrate a certain aspects of heat

DIAGRAM

The aspect of demonstrated was

- A. Conduction in different solids.
- B. Convection in water.
- C. Conduction in water.
- D. Expansion in different solids.

23. To save a person who has touched an exposed electric wire, one should

- A. Push the person away using a dry wood.
- B. Pull the person from the wire.
- C. Pour cold water on the person.
- D. Shake the person vigorously.

24. In a certain investigation pupils heated different materials and record their finding.

Which one of the following correctly shows the pupils' finding?

Materials that felt hot quickly	Materials that not feel hot quickly
A. Rubber	Iron.
B. Copper	Aluminum.
C. Plastic	Glass.
D. Steel	wood.

25. Which of the following statements is **NOT**

TRUE about friction? Friction is a force that

- A. Make work difficult.

- B. Is reduced by treading tyres.
- C. Can be used to produce fire.
- D. Affects the speed of moving objects.

26. Which of the following statements describe a human canine tooth?

- A. Chisel shaped with one root.
- B. Rough surface with two roots.
- C. Rough surface with three roots.
- D. Pointed tip with one root.

27. Which one of the following consists only of parts of human body where a digestion take place?

- A. Mouth and stomach.
- B. Large intestine and mouth.
- C. Liver and stomach.
- D. Large intestine and liver.

28. Which one of the following statements about reproduction in human beings is **CORRECT**?

- A. Fertilization take place in the uterus.
- B. Ovulation always take place twice in a month.
- C. Uterine wall construct during birth.
- D. The embryo develops into a zygote.

29. In which of the following activities is water used for recreation

- A. Watering.
- B. Cooking.
- C. Washing clothes.
- D. Swimming.

30. In which of the following is water used sparingly?

- A. Using domestic waste water to clean toilets
- B. Turning off water taps after use
- C. Recycling waste water from factories

D. Using harvested rain water for animals to drink

31. When constructing a wind vane the

- A. Pointer should be fixed on a short stand.
- B. Pointer should be firmly held to stand.
- C. Pointer should be made of cardboard.
- D. Tail of the pointer should be Large than the arrow head.

32. Which of the following groups of planets are in the correct order in terms of distance from the sun

- A. Venus, mars, earth
- B. Earth, Saturn, Jupiter
- C. Jupiter, Saturn, Uranus
- D. Mars, Neptune, Uranus.

33. The diagram below represents a weather chart prepaid by pupil

Which of the following statements is **CORRECT** about the weather chart?

DIAGRAM

- A. Whenever it was sunny it was also calm at the same time.
- B. It only rained after calm condition.
- C. Whenever it was rain it was also windy at the same time.
- D. It rained only in the afternoon.

34. Birds that have short, straight and thick beaker are

- A. Filter feeders.
- B. Flesh eaters.
- C. Nectar eater.
- D. Grain eater.

35. In which of the following does the load move through the same distance as the effort when use?

- A. Crow bar.
- B. Flag post.
- C. Spade.
- D. Claw hammer.

36. In which of the following pairs of objects will float on water?

- A. Coin and candle wax.
- B. Bottle opener and needle.
- C. Biro pen casing and rubber band.
- D. Nail and bottle.

37. During a demonstration to investigate a certain property of matter, soil was put in a small container and then dipped in water in a basin.

This was to show the presence of

- A. Air in water.
- B. Water in soil.
- A. Minerals in soil.
- B. Air in soil.

38. Which of the following pairs of materials are good conductor of electricity?

- A. Steel wool and glass.
- B. Coin and chalk.
- C. Pins and dry wood.
- D. Aluminum foil and razor blade.

39. Which one of the following statements about energy is TRUE? Energy is

- A. A push or pull.
- B. Ability to do work.
- C. Force that moves a load.
- D. Force overcome by a machine.

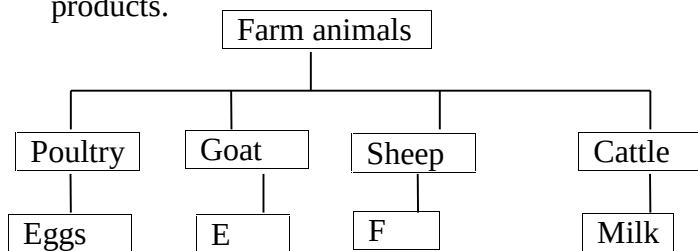
40. Driving a motor vehicle with a well maintained engine will mainly control pollution of?

- A. Air only.
- B. Water only.
- C. Soil and water.
- D. Air and water.

41. A flower which has long feathery stigma and loosely held anthers is also likely to

- A. Have brightly coloured petals.
- B. Have sweet scent.
- C. Produce many pollen grains.
- D. Produced nectar.

42. The chart shows some farm animals and their products.



Which products are represented by E and F?

E F

- | | |
|---------|---------|
| A. Milk | wool. |
| B. Wool | mutton. |
| C. Beef | mutton. |
| D. Milk | beef. |

43. The diagram below shows a set up used by pupils to carry out an investigation

DIAGRAM

The investigation was to

- A. Show the length of springs used affects balancing of different materials.
- B. Measure amount of force of different materials.
- C. Show how different materials move up and down.
- D. Compare the masses of different materials.

44. Which of the following is TRUE about soil that has high water retention? It has

- A. Large air space.
- B. Low capillarity.
- C. Fine texture.
- D. Large particles.

45. Which one of the following is an effect of air pollution?

- A. Formation of acid rain.
- B. Death of small animals.
- C. Interferes with photosynthesis.
- D. Caused respiratory diseases.

46. Which of the following groups consist only of external livestock parasite?

- A. Fleas, mites, ticks.
- B. Rounded worms, lice, mites.
- C. Fleas, tapeworms, lice.
- D. Mites, roundworms, ticks.

47. In which one of the following cases is friction a disadvantage.

- A. Grinding maize.
- B. Writing.
- C. A bird flying.
- D. Braking.

48. Which one of the following is a source of electricity?

- A. Electromagnetic.
- B. Bulb.
- C. Dynamo.
- D. Dam.

49. Which one of the following are both reasons for lighting a house?

- A. Safety and discouraging pests.
- B. Drying and discouraging pests.
- C. Warming and seeing clearly.
- D. Safety and warming.

50. In a certain investigation on soil, pupils heated soil in a container until smoke was produced. This was to investigate presence of

- A. Air in the soil.
- B. Water in the soil.
- C. Living organism in the soil.
- D. Living organisms in soil.